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Candidate interneurons mediating the resetting of the locomotor rhythm by extensor group I afferents in the cat

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ABBREVIATIONS

CDPs, cord dorsum potentials

coSur, contralateral sural

CPG, central pattern generator

DLF, dorsolateral funiculus

EFPs, extracellular field potentials

ENGs, electroneurograms

EPSPs, excitatory postsynaptic potentials

FDHL, flexor digitorum and hallucis longus

FRA/ coFRA, flexor reflex afferents / contralateral flexor reflex afferents

gr, group

GS, gastrocnemius and soleus

L-DOPA, dihydroxyphenylalanine

LDPs, locomotor drive potential

MLR, mesencephalic locomotor region

PF, pattern-formation

Pl, plantaris

RG, rhythm-generating

SmAB, semimembranosus and anterior biceps

STA, spike-triggered averaging

TA, tibial anterior

ABSTRACT

Sensory information arising from limb movements controls the spinal locomotor circuitry to adapt the motor pattern to demands of the environment. Stimulation of extensor group (gr) I afferents during fictive locomotion in decerebrate cats prolongs the ongoing extension, and terminates ongoing flexion with an initiation of the subsequent extension, i. e. “resetting to extension”. Moreover, instead of the classical Ib non-reciprocal inhibition, stimulation of extensor group I afferents produces a polysynaptic excitation in extensor motoneurons with latencies (~3.5 – 4.0 ms) compatible with 3 interposed interneurons. We assume that some interneurons in this pathway actually belong to the rhythm-generating layer of the locomotor Central Pattern Generator (CPG), since their activity was correlated to a resetting of the rhythm. In the present work fictive locomotion was (mostly) induced by i.v. injection of nialamide followed by L-DOPA in paralyzed cats following decerebration and spinalization at C1 level. In some experiments, we extended previous observations during fictive locomotion on the emergence and locomotor state-dependence of polysynaptic excitatory postsynaptic potentials (EPSPs) from extensor gr I afferents to ankle extensor motoneurons. However, the main focus was to record location and properties of interneurons (n=62) that i) were active during the extensor phase of fictive locomotion and ii) received short-latency excitation (mono-, di- or polysynaptic) from extensor gr I afferents. We conclude that the interneurons recorded fulfill the characteristics to belong to the neuronal pathway activated by extensor gr I afferents during locomotion, and may contribute to the ‘resetting to extension’ as being part of the locomotor CPG.

INTRODUCTION

It is now generally accepted that locomotor circuits in the spinal cord are organized as “central pattern generators” (CPGs) able to produce a basic locomotor pattern even without a descending control from the brain, or an afferent sensory input as first described by Graham Brown (Brown TG, 1911). A few years later Graham Brown (1914) proposed the concept of “half-centers” mutually inhibiting each other, and producing flexion and extension alternation. Further experimental evidence for this hypothesis had to await experiments by Anders Lundberg and coworkers (Jankowska et al., 1967 a,b) providing evidence for a mutual inhibition between “half-centers” activated by a particular set of “long latency flexor reflex pathways” uncovered following injection of L-DOPA (dihydroxyphenylalanine, a precursor to the monoamines dopamine and noradrenaline) in paralyzed cats following decerebration and spinalization. In this preparation stimulation of the “flexor reflex afferents” (FRA) could also evoke episodes of locomotor activity, monitored by recordings from hindlimb nerves (see also Lundberg, 1981).

Locomotion on a treadmill can be evoked in decerebrate cats i) either by stimulation of the Mesencephalic Locomotor Region (MLR) (Shik et al., 1966 a, b), or, ii) by i.v. administration of clonidine, an alpha-2 adrenergic receptor agonist following an additional acute spinalization (Forssberg and Grillner, 1973). When investigating the organization of the spinal locomotor network, it is often essential to avoid any phasic sensory feedback that could interfere with the intrinsic activity of the central locomotor network. To this end, animals are paralyzed (curarized) and the locomotor activity, i.e. “fictive locomotion”, is recorded from dissected peripheral nerves. Fictive locomotion can be evoked either by stimulation of the MLR (Jordan et al., 1979) or by the injection of L-DOPA and the monoamine oxidase inhibitor nialamide following acute spinalization (Grillner and Zangger, 1979). The pattern of efferent activity among the extensor or flexor nerves during fictive locomotion (Grillner and Zangger, 1975, 1979; Perret and Cabelguen, 1980) is just as differentiated as during real locomotion in the intact cat (Engberg and Lundberg, 1969; Rasmussen et al., 1978). There may also be occasional “deletions” of activation of a single motor nerve in some step cycles during fictive locomotion (Lafreniere-Roula and McCrea, 2005). Both of these observations are not easy to explain with a “single-layered” half-center model (Grillner and Zangger, 1975, 1979; Perret and Cabelguen, 1980, Rybak et al., 2006 a,b). Therefore the more recent models include a two-, or three-levels CPG with a rhythm-generating (RG) and a

pattern-formation (PF) level as outlined in Fig. 1A (see Burke et al., 2001, Rybak et al., 2006 a,b; McCrea and Rybak, 2007; 2008; Rybak et al., 2015; Dougherty and Ha, 2019).

Sensory input can profoundly modulate rhythmic motor activity, such as locomotion and scratching. Hip position (signaled by hip flexor proprioceptors) determined the onset of flexion during treadmill locomotion in chronic spinal cats (Grillner and Rossignol., 1978). Loading of ankle extensor muscles during stance (effectively activating Ib afferents) prolongs the extensor activity and prevents initiation of the flexor phase in decerebrate cats walking on a treadmill (Duysens and Pearson., 1980). During fictive locomotion in the cat electrical stimulation of extensor group (gr) I muscle afferents causes resetting of the locomotor rhythm by either prolongation of the extensor phase, or by termination of the flexor phase with an initiation of a new extensor phase (i.e. resetting to extension) (Conway et al., 1987). Stimulation of the same afferents also give rise to polysynaptic (with latencies of 3.5 – 4.0 ms) gr I EPSPs in extensor motoneurons (Gossard et al., 1994). The latencies of 3.5 – 4.0 ms measured from intracellular motoneuron recordings suggest at least 3 interneurons interposed in these pathways, as depicted in Fig. 1B.

Figure 1 near here

Fig. 1B outlines the neuronal circuitries that are mediating inhibitory and excitatory actions of extensor gr I muscle afferents during locomotor activity. The uppermost level is referred to as the “intermediate nucleus”, and corresponds to the region where Eccles et al. (1954) described the largest monosynaptic muscle gr I fields for the first-order interneurons. This seems to correspond to the medial part of Rexed’s lamina VI (Rexed, 1954; 1964). The scheme of the locomotor CPG elements in Fig. 1 depict the intermediate nucleus as the location of the first-order “input interneurons” for the extensor gr I excitatory pathway, rather than being part of the RG interneurons. What is the basis of this viewpoint? The early experiments tracing the monosynaptic terminal fields from the reticulospinal pathways relaying the MLR locomotor signal to the lumbar spinal cord during MLR-evoked locomotion (Noga et al., 1995) suggest that they make excitatory connections with interneurons preferentially in lamina VII. A second key evidence is the work by Kjaerulff and Kiehn (1996). They observed that the locomotor activity induced by 5-HT+NMDA

in the isolated (*in vitro*) preparation of the neonatal rat spinal cord, remained after removing the dorsal aspect of the spinal cord by a horizontal section, provided this did not reach below the level of the central canal. This certainly suggests that interneurons ventral to the intermediate nucleus, i.e. in lamina VII, is sufficient to support the rhythm-generating capability, but it does not exclude that interneurons in the intermediate nucleus also participate in the rhythm-generation.

In the absence of locomotor activity, the activation of extensor gr I muscle afferents evokes the classical “autogenic Ib inhibition” of motoneurons to homonymous and synergic muscles as originally described by Eccles et al. (1957). This inhibition, which is attenuated during locomotion, is mediated by pathway *a*, in Fig. 1B. Following later investigations (Harrison et al., 1983; Harrison and Jankowska, 1985 a, b) revealing a much broader afferent input to interneurons mediating the autogenic Ib inhibition, as well as a wider distribution of interneuron outputs, this pathway was then referred to as “gr I non-reciprocal inhibition”. During MLR- or L-DOPA-evoked locomotion, there is a “reversal” of the extensor gr I inhibition during rest to a dominating excitation during locomotion (Schomburg and Behrends, 1978; Gossard et al., 1994; McCrea et al., 1995; Guertin et al., 1995; Angel et al., 1996; Angel et al., 2005; see also Pearson and Collins, 1993). Pathway *b* in Fig. 1B is mediating the locomotor-related disynaptic excitation emerging via one interposed interneuron (Schomburg and Behrends, 1978; McCrea et al., 1995; Guertin et al., 1995; Angel et al., 1996; Angel et al., 2005). Candidate interneurons of this pool have been located in the L7 segment, receive monosynaptic input from gr I muscle afferents, and are last-order neurons projecting to ipsilateral ankle extensor motoneurons (Angel et al., 2005).

The pathway of the polysynaptic gr I excitation (with latencies of 3.5 – 4.0 ms) is denoted by *c* in Fig. 1B. During fictive locomotion, stimulation of extensor gr I afferents during flexion resets the step cycle to extension (Conway et al., 1987), as well as during fictive scratch (Perreault et al., 1999). Moreover, during fictive locomotion or scratch, it is the polysynaptic EPSPs that dominate in extensor motoneurons (Gossard et al., 1994). The interneurons mediating polysynaptic EPSPs of extensor motoneurons are to a great extent located in the L6-L7 segments since these polysynaptic EPSPs remain following a spinal transection at L6 level (Gossard et al., 1994). These findings led Gossard et al. (1994) to suggest that the polysynaptic extensor gr I EPSPs were actually mediated by the rhythm-generating interneurons.

The main aim of this study is to examine in the cat spinal cord the activity and location of candidate interneurons mediating the polysynaptic extensor gr I excitatory pathway (during fictive locomotion (and partly fictive scratching), i.e. the pathway depicted with *c* in Fig. 1B. These interneurons are also candidates to mediate the resetting to extension of the locomotor step cycle and thus part of the rhythm generation layer of the CPG. During the early part of this series of experiments, we extended previous observations on inhibitory and excitatory postsynaptic potentials from extensor gr I afferents to ankle extensor motoneurons during fictive locomotion.

Preliminary reports have been published earlier (Stecina et al., 2007, Domínguez-Rodríguez et al., 2016; 2018).

METHODS

The results presented in this study were obtained from 42 cats of either sex, weighing between 2.5 – 4.7 kg (n=4 at Copenhagen University, Denmark (*CPH*); n=38 at CINVESTAV-IPN, Mexico City (*MC*). All surgical and experimental procedures were conducted in accordance with EU regulations (Council Directive 86/609/EEC) and the National Institutes of Health Guidelines for the Care and Use of Laboratory Animals (USA). The protocols were approved by the Danish Animal Experimentation Inspectorate (for experiments in Denmark) and by the Institutional Committee for the Use of Animals in the “Centro de Investigación y de Estudios Avanzados, CINVESTAV” (for experiments in Mexico). When differences in Methods were present in the Danish and Mexican parts of the project, we will specify that by reference to *CPH* and *MC*, respectively.

Anesthesia and surgery. Anesthesia was induced by halothane–nitrous oxide (2–3% halothane, 70% N₂O, and 30% O₂) (*CPH*), or isoflurane (1.5% isoflurane in O₂) (*MC*). After a tracheotomy and intubation, anesthesia was maintained throughout the surgery with either halothane (0.8–1.5%) delivered in an oxygenated mixture of nitrous oxide (60% N₂O, 40% O₂) (*CPH*), or isoflurane (0.8–1.2%) delivered in O₂ (*MC*). Cannulas were inserted into the right carotid artery and the right jugular vein (near the trachea) for monitoring of blood pressure and the administration of fluid and drugs. At the beginning of the surgery, the following drugs were administered: atropine (0.1 mg kg⁻¹, s.c.) to control excessive bronchial secretions, and

dexamethasone (1.0 mg/kg, i.v.) to prevent edema. A buffer solution (10% dextrose and 1.7% NaHCO₃) was infused continuously (2.5 - 4.5 ml h⁻¹ i.v) after cannulation.

Hindlimb nerve dissection included the left (ipsilateral – in relation to the recorded motoneurons and interneurons) posterior biceps and semitendinosus, *iPBSt*; semimembranosus and anterior biceps, *iSmAB*; gastrocnemius and soleus, *iGS*; plantaris, *iPl*; flexor digitorum and hallucis longus, *iFDHL*; tibial anterior, *iTA*; extensor digitorum longus, *iEDL*; sural, *iSur* and superficial peroneal, *iSP*. In some cases the quadriceps (*iQ*) and sartorius (*iSart*) nerves were also dissected and later mounted for stimulation and recording. On the right (contralateral) side the *coGS*, *coTA*, *coSur* and *coSP* nerves were dissected and later mounted on electrodes.

As described in the Introduction, loading of extensor muscles (activating the extensor gr I afferents) may delay the swing phase, or even result in a maintained extension. Also, extension of the hip determines the onset of the subsequent flexor phase. Even a tonic activation of these (and other) afferents may affect the rhythm and pattern of fictive locomotion. Therefore all branches of the major hindlimb nerves (femoral, obturator and sciatic) were sectioned bilaterally in order to remove tonic afferent input from the hindlimbs during fictive locomotion /scratching.

Laminectomy was performed and the dorsal aspects of vertebrae at the L1-L7 levels were removed. After these surgical procedures, the animal was transferred to a stereotaxic frame (still under halothane/isoflurane anesthesia). The vertebral column was fixed by means of spinal and pelvic clamps and the head by means of ear bars, an eye pit and upper jaw holders. Pools were made with the skin flaps around the exposed tissues, filled with mineral oil and warmed by radiant heat. The temperature was monitored by one thermometer inserted into the esophagus (*MC*) or rectum (*CPH*) and a second one placed in the back pool; both body temperature and the oil pool were maintained at 37 °C.

Following a craniotomy, a precollicular-postmammillary decerebration was performed with removal of the rostral parts, including the cerebral cortex, allowing the anesthesia to be discontinued. The animal was subsequently paralyzed with pancuronium bromide (0.2 mg/kg, supplemented every 40–60 min) and artificially ventilated to maintain end tidal pCO₂ at 3–5 %. A bilateral pneumothorax was performed to minimize respiratory movements and thus stabilize microelectrode recordings in the spinal tissue. At the end of each experiment, animals were euthanized with an overdose of pentobarbital.

Electrophysiological stimulation and recording. The nerves were mounted on bipolar hook silver electrodes placed at the distal end of the dissected and sectioned nerves for recording (electroneurograms; ENGs) and stimulation. In a minority of the experiments, the *iQ* and *iSart* nerves were similarly placed in bipolar silver wire cuff electrodes. We recorded the cord dorsum potentials (CDPs) by a silver ball electrode placed on the surface of the ipsilateral dorsal column at L7 (near the dorsal root entry zone) with a reference electrode on the back muscles. We were thus able to record the afferent incoming volleys evoked by the stimulation of peripheral nerves with single pulses of 200 μ s or train of pulses at 200 - 300 Hz, and with a strength twice the threshold (xT) required to activate the most excitable fibers (n.b. 2 xT to maximally activate gr I afferents).

A tungsten electrode (0.1M Ω , for either stimulation or recording) was placed at L7 level in the ventral horn through the dorsolateral funiculus (DLF) with an angle of 20°. In order to localize extensor motor nuclei we recorded the antidromic extracellular field potentials (EFP) evoked by stimulation of extensor nerves (Pl, GS). Then the motor nucleus was subsequently stimulated with graded strengths (< 5-20 μ A) and the ENG from extensor nerves were recorded. The threshold for evoking a direct activation of motor axons should be < 5 μ A.

MLR-induced fictive locomotion. Cats (n=8) were decerebrated by a precollicular-postmammillary transection as described above. An insulated, monopolar tungsten electrode was lowered into the MLR (Horsley-Clarke coordinates P2, L4, H0; Shik et al., 1966 a, b; Jordan et al., 1979). The position of the electrode was adjusted to obtain stable fictive locomotion. The stimulus parameters were usually 30-150 μ A, 0.5-1.0 ms rectangular pulses delivered at 10-30 Hz. The stimulation was usually unilateral (ipsi- or contralateral). Additionally in these preparations fictive scratching was induced in the left hindlimb by manual stimulation of the left pinna, subsequent to topical application of D-tubocurarine (0.1–0.5%) at C1 and C2 dorsal root entry zones (Domer and Feldberg, 1960).

L-DOPA induced fictive locomotion. Decerebrated cats (n=34) were spinalized at high cervical (C1) (following intraspinal administration of lidocaine) to block tonic descending inputs to the

spinal circuits. Fictive locomotion was induced by pre-treatment with nialamide (50 mg/kg i.v., given slowly 2-3 hours before spinal transection), followed by L-DOPA (50-100 mg/kg i.v) administration at about 1-2 hours after spinalization (Jankowska et al., 1967a, b; Viala and Buser, 1969; Grillner and Zangger, 1979; Conway et al., 1987; Gossard et al., 1994). When the L-DOPA-evoked fictive locomotion failed after several hours, i.v. injection of 4-aminopyridine (0.25-1.0 mg/kg) (Zangger, 1981; Dubuc et al. 1986), or naloxone (0.2–1.0 mg/kg; Pearson et al., 1992; Schomburg and Steffens, 1995) sometimes could restore the locomotor activity. Brief high strength stimulus trains to *co*FRA or the contralateral dorsolateral funiculi (DLF, at L3 level) could also trigger locomotion for shorter periods (minutes). Also in this preparation (with spinalization at C1) it was sometimes possible to trigger scratching subsequent to topical application of D-tubocurarine at the C1 and C2 dorsal root entry zones and by tactile stimulation of left or right pinna.

Intracellular recording from motoneurons. All recorded motoneurons (n=12 cats) were antidromically identified as extensor motoneurons by the stimulation of ipsilateral extensor nerves (*iGS*, *iPl*, *iFDHL* and *iSmAB*). Intracellular recordings were performed with glass micropipettes (3-5 M Ω) filled with 2M K⁺ acetate and 100 mM of QX-314 to block action potentials. Micropipettes entered from the dorsolateral funiculus at caudal L6-L7 level with an angle of around 20°. Orthodromic EPSPs evoked by stimulation of extensor nerves were recorded during L-DOPA-induced fictive locomotion.

Extracellular recording from interneurons.

Orthodromic extracellular field potentials (EFPs) evoked by the stimulation of ipsilateral extensor nerves and extracellular interneuron spikes rhythmically activated during locomotion were recorded (n =30 cats) with glass microelectrodes (3-7 M Ω) filled with Na⁺ glutamate (3 M). To increase interneuronal firing negative current was applied to eject glutamate. Firing of a single interneuron was used for spike-triggered averaging (STA; see below) of field potentials recorded in the motor nuclei, and ENGs recorded in peripheral nerves, to determine the projection of the interneuron (i.e. identifying last-order interneurons). The “first-order” interneurons are those

interneurons in reflex pathways, which receive direct monosynaptic excitation from primary sensory afferents. The “last-order” interneurons are those that are projecting to the motoneurons. For a disynaptic reflex pathway the interposed interneurons are thus both “first-order” and “last-order”.

Micropipettes entered through the lateral part of the ipsilateral dorsal column at L5-L7 with an angle around 0° to 10° medial. We searched for interneurons which were 1) active during the extensor phase of locomotion (or scratch), and 2) activated by brief trains of gr I stimulation of ipsilateral extensor nerves (mono-, - di- or polysynaptically). Interneurons were included in the material if they were not antidromically activated by an electrode placed in the contralateral intermediate nucleus or ventral horn (excluding commissural neurons). We also excluded ascending tract neurons that were excited by stimulation with electrodes placed bilaterally at the dorsolateral funiculi (DLF) at L1 level.

Intracellular recordings from motoneurons and extracellular recordings of interneurons were performed in bridge mode using an Axoclamp-2B amplifier (Molecular Devices, USA; bandwidth from DC to 10 kHz, and a sampling frequency of 30 kHz). ENG recordings of peripheral nerves monitored locomotor activity by means of custom-made AC differential amplifiers (30 Hz - 3 kHz bandwidth) from the Department of Neuroscience (*CPH*), or the Spinal Cord Research Center, SCRC, University of Manitoba, Canada (*MC*). Raw ENG recordings of peripheral nerves were collected in gap-free mode protocol using a Digidata 1440A card and the AxoClamp software v.10 (Molecular Devices, USA) (*MC*), or the 1401 analog-to-digital converter and the Spike2 software (Cambridge Electronic Design, Cambridge, UK) (*CPH*).

Spike-triggered averaging (STA) analysis. We identified last-order interneurons by antidromic activation from extensor nuclei (mainly the GS motor nucleus). In order to elucidate whether last-order interneurons were excitatory, we recorded responses in the rectified extensor ENGs that were the result of last-order interneuron spontaneous firing during fictive locomotion or scratch. The excitatory action by the interneuron onto the extensor motoneurons was reflected as the appearance of a peak in the averaged extensor ENG activity following the triggered spikes (see Fig 6D, cf Fetz et al., 1996). We performed the STA analysis and measured the latency from the interneuron spike to the onset of the ENG peak (Analysis Software, SCRC, University of

Manitoba, Canada). To determine whether this latency corresponded to monosynaptic activity we compared it with the conduction time measured from the corresponding afferent volley (correcting for the slower conduction velocity in the motor axons than in the fastest afferents).

RESULTS

Intracellular recordings from ankle extensor motoneurons

Gossard et al. (1994) illustrated how the gr I non-reciprocal inhibition seen at rest virtually “disappeared” and was replaced by a large polysynaptic gr I EPSP as locomotion was initiated by either L-DOPA injection (their Fig. 1) or MLR stimulation (their Fig. 4). However, the overwhelming excitation usually left a small but still significant early inhibition (the gr I non-reciprocal inhibition) followed by the longer latency excitation. Here we have confirmed the presence of this early inhibition during fictive locomotion (Fig. 2 Bi and Bii), though certainly “overshadowed” by the subsequent excitation, especially following a train of stimuli. The amplitude of the polysynaptic gr I excitation in extensor motoneurons depended on the phase of the fictive step cycle in which the gr I stimulus train is given. In the GS-motoneuron illustrated in Fig. 2, the EPSPs evoked during the depolarized (extensor) phase (expanded time course in the blue vertical traces in Fig. 2A) were much smaller than the EPSPs evoked during the flexor phase (expanded time course in the red vertical traces). This is further demonstrated in Fig. 2Bi and Bii for the average of gr I EPSPs during the flexor (red) and extensor (blue) phases respectively. Extensor gr I polysynaptic EPSPs evoked by a train of stimuli during the flexor phase had an amplitude that, at most, equaled the amplitude of the depolarized active phase of the locomotor drive potential (LDP), as illustrated in Fig. 2A.

Figure 2 near here

The decrease of the gr I EPSP amplitude during the extensor phase may be explained by an occlusion that would be expected if the depolarization from the locomotor centers and the gr I EPSPs were transmitted by the same interneuron pools. The smaller amplitude during the extensor

phase is not due to “shunting”, i.e. an increased conductance. Previous work demonstrated an increased conductance during locomotion and scratching in relation to the preceding rest, but no further increase during the depolarizing phase of the LDPs (for locomotion, Shefchyk and Jordan, 1985; for scratching, Perreault, 2002). In the present experiments, we have confirmed these observations in a few motoneurons by comparing the input resistance (by short current pulses) during the flexor and extensor phases, respectively.

Note that gr I EPSPs seem particularly large when the stimulus trains occurred at the transition from flexion to extension as if the transmission had been facilitated (marked by the 3 arrows and 1 star in Fig. 2A). It also appears that brief gr I stimulus trains are capable to trigger a faster transition to the extensor phase in Fig. 2A (see arrows corresponding to the 1st, 2nd and 4th transitions to extensor activity). With the 3rd transition (marked by a star) the stimulus train occurred so early (before the spontaneous transition) so that the membrane potential transiently fall back to the hyperpolarized level, and it appears that the transition to the extensor phase was slower. Indeed, when we measured the duration of the rising phase of the LDP without the gr I trains (control; exemplified by upper trace in Fig. 2C) it is slower than the rising phase of the LDP after the short train of stimuli to Pl gr I afferents (lower trace in Fig. 2C). In order to confirm that this difference in rising phase duration is significant, we performed t-test analysis comparing the duration of all the rising phases of the LDPs (for control locomotor episodes of 8 step cycles without stimulation, and the 12 step cycles with stimulation of gr I Pl afferents occurring just before the spontaneous transition). The p-value of the t-test was 0.0004, and thus very significant. The early gr I non-reciprocal inhibition was regularly seen as illustrated in Fig. 2B. It was evident with single pulse stimuli, as it was not “overshadowed” by the larger EPSPs evoked with trains of stimuli (compare Fig. 2Bi and Bii). The amplitude of the early gr I IPSP was larger during the active (depolarized) phase (see the averaged and superimposed records in Fig. 2 Bi and Bii). It is not possible to conclude whether this depends on a facilitation of the interposed interneuron during extension, or it was due to the depolarization itself (i.e. driving force between the level of the depolarized E_m and the equilibrium potential of Cl^-). The appearance of a gr I disynaptic EPSPs in a GS motoneuron during the extensor phase of locomotion is illustrated in Fig. 3. The layout is similar to Fig. 2 and illustrates distinct responses to extensor nerve stimulation (GS 2xT, single pulse) from a motoneuron recorded intracellularly during a period of

fictive locomotion (A) and the averages during flexion (B) and during extension (C). The monosynaptic Ia EPSPs dominate the response, but during the active (extensor) phase there is also a disynaptic component. We found such disynaptic EPSPs 10 of 39 recorded motoneurons, and they were always facilitated during the extensor phase.

Figure 3 near here

Candidate interneurons mediating gr I evoked polysynaptic EPSPs in extensor motoneurons and enabling resetting to extension

The interneuronal pathway which is mediating gr I polysynaptic excitatory response to extensor motoneurons involves at least three interneurons interposed between gr I extensor afferent fibers and extensor motoneurons based on observations from intracellular motoneuron recordings (cf. Fig. 1B and Gossard et al., 1994). Interneurons that could belong to pathways mediating gr I excitatory response on extensor motoneurons were examined by collecting extracellular recordings from interneurons fulfilling the following criteria: i) mono-, di- and/or -/poly-synaptic activation by stimulation of extensor gr I afferents during fictive locomotion (or scratch), and ii) rhythmic bursting during extensor phase of fictive locomotion (or scratch). Interneurons with antidromic activation by bilateral stimulation from the dorsolateral funiculi (DLF) at L1 level (ascending sensory neurons) and those with antidromic activation from the contralateral intermediate nucleus (commissural interneurons) were excluded, to focus primarily on interneurons possibly involved in flexion / extension alternation of the ipsilateral limb. Input from gr I afferents from all ipsilateral mounted nerves (from about 6-8 individual muscles, see Methods) were tested and then we recorded the interneuronal activity during resetting of the locomotor or scratching rhythm as trains of stimulation to extensor gr I afferents or trains to contralateral flexor reflex afferents (*coFRA*) were delivered. Parallel changes in the neural discharges of the recorded interneuron and of extensor nerve activity (ENGs) was expected. When “resetting to extension” with stimuli given during the flexor activity, we expected both the interneuronal activity and the extensor ENGs display a shift in the onset of the subsequent burst (i.e. an earlier onset). When the stimuli were given during the extensor phase, we expected a prolongation of the ongoing extensor phase with a postponed subsequent flexor phase, in addition

to the enhancement of extensor ENG activity. We tested whether the interneuron projected to extensor motoneurons by testing antidromic activation when low threshold stimulation of the *iGS* motor nucleus was delivered. In the following sections, we will present 4 examples of interneurons (in Figs 4-7) activated by extensor gr I afferents at different latencies fulfilling the criteria of being candidates for neurons in pathways enabling resetting of the step cycle.

Figure 4 near here

Extracellular recording from an interneuron located at the rostral L6 segment, medially in the intermediate nucleus at a depth of 2.5 mm (see Fig. 8C for the location) is activated by stimulus trains to *iFDHL* nerve at 2 xT with an onset latency of 0.9 ms (Fig 4A). The spikes are superimposed on the monosynaptic gr I field potential in the intermediate nucleus and the action potentials show small variations in onset latency as would be expected with a monosynaptic connection. This interneuron was rhythmically firing in phase with extensor ENG activity (in the GS and FDHL nerves, see Fig. 4B) during L-DOPA-induced locomotion. After a stimulus train to *coFRA*, the interneuron fired with a delay similar to the onset of the long-latency crossed extensor burst recorded from the *iGS* nerve as shown in Fig. 4D. All of these data together suggest that this interneuron could be one of the “first-order” neurons of the gr I polysynaptic excitation onto extensor motoneurons during fictive locomotion, as outlined in Fig. 4C (see further in the Discussion).

The interneuron shown in Fig. 5 was recorded at a depth of 2.3 mm but rather medial (see Fig. 8C). It displayed action potentials at an onset latency of 1.3 ms as measured from the *iGS* incoming volley (2 xT stimulation; Fig. 5A) suggesting a disynaptic linkage. There was also a disynaptic excitation from *iSmAB* (not illustrated). Thus, it could be a “second-order” interneuron in the resetting pathway as outlined in Fig. 5C. The interneuron was rhythmically activated during the extensor phase of L-DOPA evoked locomotion (Fig. 5B) and high frequency bursting was evident when the extensor ENG amplitude was largest while the firing frequency was lower during burst with smaller *iGS* ENG amplitude like the 3rd extensor phase shown in Fig. 5B. This interneuron was also excited by *coFRA* stimulation (Fig 5D; *coSur* 50 xT) in synchrony with the activation of the *iFDHL* ENG. Furthermore, the action potentials evoked by extensor gr I afferent

stimulation (Pl 2 xT) during flexion show the recruitment of this neuron during resetting to extension (Fig. 5E).

Figure 5 near here

The illustrated cells in Fig. 6 and 7 are both last-order interneurons, i.e. projecting directly to motoneurons, as judged from their antidromic activation by low threshold stimulation in the GS motor nucleus. The interneuron in Fig. 6 (caudal L6 segment, medial in the ventral horn at a depth of 4.2 mm; see Fig. 8C) was activated by Pl, SmAB and GS (not illustrated) at a disynaptic latency (Fig. 6 Ai and Aii). Even though the shortest latency was clearly disynaptic, the rather large variation in latencies (1.3 ms – 2.8 ms) suggested that there were also longer-latency (trisynaptic) excitation contributing. We have therefore classified this interneuron as receiving a combination of di- and polysynaptic input (orange circle in Fig. 8C). During L-DOPA-evoked locomotion this interneuron fired simultaneously with the activity of the extensor nerves (SmAB and Pl). Note, however, that the SmAB nerve likely contained a branch from PB causing it to be active also during the flexor phase (together with TA). Action potentials in this neuron were present only with the most intense activity of the extensors (Fig. 6B). Stimulation in the GS motor nucleus (15 μ A) evoked antidromic discharges (Fig. 6C). Moreover, spike-triggered averaging (STA; n=1450) revealed an evoked excitatory response in the SmAB nerve with a latency of 1.5 ms. This delay fits well with a short intraspinal projection from the recorded interneuron, the synaptic delay from its presynaptic terminals onto SmAB motoneurons and the conduction time to the SmAB nerve. In the sketch in Fig. 6E we have suggested that this interneuron is both a 2nd and 3rd order interneuron in addition to its direct projection to SmAB motoneurons.

Figure 6 near here

The interneuron illustrated in Fig. 7 was activated only at polysynaptic latencies from gr I trains to GS and SmAB nerves (Fig. 7A). This interneuron was recorded from the caudal L6 segment, in the ventral horn at a depth of 3.4 mm; see Fig. 8C. It fired in synchrony with the FDHL nerve (Fig. 7B). The activity in the FDHL nerve seems to have a “double-peak” in the

pattern of activity during locomotion, most likely because the digitorum and hallucis longii muscles have different activity patterns during locomotion (see O'Donovan et al., 1982, and Fleshman et al., 1984). This interneuron was antidromically activated from the GS motor nucleus (Fig. 7C) and therefore likely a last-order cell as shown in Fig. 7D.

Figure 7 near here

We have collected data from interneurons during both L-DOPA-evoked locomotion in spinal cats (as illustrated in Figs. 4-7; 45 interneurons), and MLR-evoked locomotion in cats with intact spinal cords (8 interneurons). In cats with L-DOPA-induced locomotion we recorded only 2 interneurons during fictive scratch. In non-spinalized cat intended for MLR stimulation we recorded interneuron activity during scratch (11 interneurons; 4 of them during both scratch and locomotion).

Figure 8 near here

Figure 8A illustrates the position of all 62 interneurons fulfilling the criteria of being activated by extensor gr I afferents, and being active during the extensor phase of fictive locomotion (and/or fictive scratch). Most interneurons (58/62) were recorded from the L6 to the rostral L7 segments and only 4 cells in the L5 segment. The location of each interneuron was based on the recorded depth measured from the surface of the spinal cord at the entrance of the microelectrode, and on the angle of the electrode track. The depicted position for each neuron was also adjusted (within 0.5 mm) according to the recorded field potentials – especially the gr I field potentials in the intermediate nucleus (see Eccles et al., 1954; their Figs. 1 and 2). Interneurons with only mono-, di-, or tri-/polysynaptic excitation are marked with circles in blue, yellow and red color respectively. However many interneurons received excitation with several latencies and those are marked by green (with mono- and di-synaptic excitation, n=4) or orange (with a polysynaptic component, but with additional mono- and/or di-synaptic contributions from different extensor nerves). In the case of the 14 interneurons that were exclusively excited at a monosynaptic latency (blue circles), most (n=10) were located in the intermediate nucleus. Of the

8 interneurons exclusively excited at a disynaptic latency (yellow circles), 5 were located in the intermediate nucleus, and 3 in the ventral horn. The 13 interneurons with only polysynaptic excitation (red circles) were all located in the ventral horn (lamina VII), and they were preferentially localized to the lateral side. Those interneurons that were polysynaptically activated, but in addition had a mono- or disynaptic component ($n=23$; orange color in Fig. 8A) were spread over the intermediate nucleus and the ventral horn. In Fig. 8B we have marked all interneurons with a monosynaptic component, regardless of possible additional di- or polysynaptic components as well. Golgi tendon organ (Ib) afferents do not extend deeper than the intermediate field (Brown and Fyffe, 1979; Hongo et al., 1978). The interneurons in the ventral horn that were monosynaptically activated by gr I afferents thus were most likely activated by muscle spindle Ia afferents, and therefore likely be the interneurons mediating the reciprocal inhibition (see Jankowska and Lindström, 1972). They would thus not belong to the gr I excitatory pathway to extensor motoneurons.

As described in the Methods we tried to identify the last-order interneurons in two ways; i) antidromic activation from the GS motor nucleus, ii) spike-triggered averaging (STA) of evoked potentials in extensor nerves. Antidromic responses from the GS motor nucleus were seen in 6 interneurons (two of them are shown in Figs. 6 and 7). Another antidromically activated interneuron had monosynaptic excitation from GS and Pl gr I afferents, and it was located within the intermediate nucleus. At the border between the intermediate nucleus and the ventral horn, a last-order interneuron was also found with both mono- and disynaptic excitation from FDHL. They may represent interneurons mediating the disynaptic excitation, which was strongly facilitated during the extensor phase (see Fig. 3). The remaining 4 interneurons with antidromic activation from the GS motor nucleus were all located deeper in lamina VII (ventral horn), and received only polysynaptic excitation ($n=1$; Fig. 7), or a mixture of di- and polysynaptic excitation ($n=3$; one cell illustrated in Fig. 6). Evoked potentials in extensor nerves were observed by STA from 4 interneurons (exemplified in Fig. 6). All of these interneurons were also antidromically activated from the motor nucleus and all were located in the ventral horn.

DISCUSSION

We will start the discussion by setting the current project into a broader functional context. During treadmill walking in decerebrate cats, Duysens and Pearson (1980) found that an artificial loading of the ankle extensor muscles (triceps surae) of one limb enhanced the extensor EMG-activity during stance (i.e. by activation of Ib afferents) and delayed the subsequent flexor burst generation. With artificially maintained load, the swing phase (flexor burst) was cancelled and a tonic extensor activity was recorded from that limb, while the normal pattern of locomotion continued in the contralateral hindlimb. During over-ground walking, the enhancement of extensor activity, and the delay of the flexor phase could be of major importance to prevent the initiation of the subsequent swing phase at a time when hindlimb extension is fully needed to support the weight of the animal. In contrast, during the flexor phase the activity of ankle extensor Ib afferents is likely reduced or absent. However, stimulation of ankle extensor gr Ib afferents during the flexor “non-functional” phase of fictive locomotion unveiled the “resetting to extension” (Conway et al., 1987). In the Results section we have referred to “extensor gr I excitation” as it is rather impossible to distinguish between muscle spindle Ia afferents and Golgi tendon organ Ib afferents by graded electrical stimulation of the peripheral nerves. However, based on a more physiological activation of Golgi tendon organ Ib afferents by electrically evoked contraction of extensor muscles (Conway et al., 1987), and of muscle spindle afferents by tendon vibration (Conway et al., 1987; Pearson and Collins, 1993) it was concluded that the major origin of the rhythm resetting seems to originate from activation of Golgi tendon organ Ib afferents. The sensory input with access to the rhythm generator is not limited to the extensor Ib afferents. Resetting of the rhythm generation can also be achieved from low threshold cutaneous afferents (Quevedo et al., 2005 a, b), gr II afferents from flexor muscles (Stecina et al., 2005) and high threshold “flexor reflex afferents” (Jankowska et al., 1967 a,b; Gossard et al., 1994).

The more specific motivation to study the extensor-related interneurons in the present work relates to the discovery of the strictly locomotor-related expression of polysynaptic extensor gr I EPSPs in extensor motoneurons (Gossard et al., 1994), and the subsequent finding of a similar reflex pathway during scratching (Perreault et al., 1999). The latency of 3.5 – 4.0 ms suggested three interposed interneurons in the pathway. These authors hypothesized that the “three-interneuron pathway” was actually *causing* the resetting to extension, and that it was mediated by

the rhythm-generating interneurons on the extensor “half-center” as presented in Fig. 9A. It would therefore be physiologically very relevant if those interneurons could be identified.

Figure 9 near here

We provide data from 62 interneurons which fulfilled two main criteria, i) mono-, di- and/or tri-/polysynaptic activation by stimulation of extensor muscle gr I afferents during fictive locomotion or scratch, and ii) rhythmic bursting during extensor phase of fictive locomotion or scratch. In accordance with Jankowska et al. (1967 a,b) the interneurons which belong to the extensor “half-center” should be activated in response to stimulation of the *coFRA*. We have provided examples (Figs. 4D, 5D), that the *coFRA* stimulation indeed was able to activate the candidate “resetting interneurons” together with the activation of the extensor ENGs (22/22 of the tested interneurons). Gossard et al. (1994) demonstrated in intracellular recordings from extensor motoneurons that there is a convergence at interneuronal level from extensor gr I afferents and *coFRA*, i.e. a facilitation of the polysynaptic gr I EPSPs when conditioned by the *coFRA*; confirmed in the present investigation. Thus, the excitatory interneuronal pathway mediating the gr I polysynaptic EPSPs and the one which mediates long-latency *coFRA* L-DOPA reflexes share common elements, probably related to the extensor “half-center” circuitry.

Our results demonstrate that the intermediate nucleus contains several interneurons with only monosynaptic excitation from extensor gr I afferents, however, the majority of the interneurons in the intermediate nucleus had a convergence of mono- and disynaptic inputs. Some had only disynaptic excitation, and several had a mixture of mono- di- and polysynaptic excitation (summarized in Fig. 8A). The rhythm generator layer of the CPG is thought to be performing the “clock function”. The pattern formation network contains multiple interneuron populations each of which projects to synergist motoneuron populations to achieve the specific pattern of muscle activity during locomotion (Rybak et al 2006a; see also our Introduction).

The explanation for the mixture of mono- di- and polysynaptically activated interneurons suggest a feed-back from the rhythm- and pattern-generating layers to the intermediate nucleus to adjust transmission through a large number of spinal pathways. These would not only include the extensor gr I pathways denoted *a*, *b* and *c* in Fig. 9, but also other reflex pathways including

cutaneous reflexes (Forssberg, 1979, see the review by McCrea, 2001) and phase-dependent modulation of the presynaptic inhibition of muscle Ia and Ib afferents primary afferents (Dueñas and Rudomin, 1988; Gossard, 1996; Gossard et al., 1991) as well as cutaneous primary afferents (Gossard et al., 1989). The locomotor-related firing of interneurons of these pathways (partly located in the intermediate nucleus) would thus reflect the *output* from the CPG and not the input to the rhythm-generator *per se*. All the interneurons with only polysynaptic excitation are located in the ventral horn (red in Fig. 8), and we have several interneurons with disynaptic excitation (yellow in Fig. 8), or with a mixture of polysynaptic and shorter-latency excitation (orange in Fig. 8). In relation to Fig. 8B, we propose that monosynaptically excited interneurons in the ventral horn most likely are examples of interneurons mediating disynaptic Ia inhibition.

Altogether, there is an extensive mixture of mono- di- and polysynaptic excitation of the sampled interneurons, which does not correspond to a strict division between the “three layers” as suggested in Fig. 1A. We need to be open-minded about the possibility that some interneurons in the intermediate nucleus are involved in the rhythm-generation. It is our hypothesis that the extensor polysynaptic gr I EPSPs to extensors are mediated by the rhythm generator (extensor half-center) as outlined in Fig. 9A, but the undisputable proof for this is still lacking. If we consider the alternative possibility that they are *not* mediated by the rhythm generator, but by another separate pathway (as in Fig. 9B), the resetting response would require that the extensor gr I afferents have an additional pathway acting on the rhythm generator, possibly by a complex, long-latency pathway (not depicted in Fig. 9B). If this were the case, then the rhythm generating interneurons must be facilitating not only the disynaptic, but also the polysynaptic gr I pathway as indicated in Fig. 9B. Actually, there has not been an accurate estimation of the latency for the resetting *per se*, and it would indeed be very difficult. It could be expected that the latency would be longer if evoked in the beginning of the flexor phase than in the end close to the spontaneous transition to the subsequent extensor phase. Nevertheless, the observations in Fig. 2C strongly suggests that a short train (5 pulses) actually could speed up the otherwise spontaneous switch from flexion to extension, and this supports the hypothesis that the 3.5 – 4.0 ms excitatory pathway is actually mediated via the rhythm generating circuit as outlined in Fig. 9A.

Traditional anatomical and electrophysiological methods have only been conclusive for analysis of disynaptic pathways with only one interneuron interposed between the input and the

target motoneuron. The examples include the identification of the Ia inhibitory interneurons (reciprocal inhibition) (Jankowska and Roberts, 1972; Jankowska and Lindström, 1972; Brink et al., 1983), the Renshaw cells (Brink et al., 1983; Moore et al., 2015) and the interneurons mediating the disynaptic gr I non-reciprocal inhibition (Brink et al., 1983). It would be virtually impossible to track the “three-interneuron polysynaptic pathway” by classical electrophysiological and anatomical methods, and other approaches are required for disclosing these more complex pathways. For example, by recording from candidate rhythm generating interneurons marked by specific developmental transcription factors and using molecular genetic tools could be one approach in mouse preparations (see reviews by Jessell, 2000, and Goulding 2009; Kiehn et al., 2010; Gosgnach et al., 2017; Dougherty and Ha, 2019). There are a more limited number of research publications of more direct significance to our present work, and that include the cholinergic V0_C interneurons (Zagoraoui et al., 2009), the Hb9 interneurons (Wilson et al. 2005 and Caldeira et al. 2017), and not least the Shox2 excitatory interneurons described by Dougherty et al. (2013). Li et al. (2019) have recently investigated whether a group of Shox2-nonV2a neurons (presumed rhythm-generating neurons; see Dougherty et al., 2013) also received short latency excitation from extensor gr I afferents in the neonatal mouse. The work by Li et al (2019) is still not conclusive with regard to the specific short-latency excitation from extensor gr I afferents, but certainly opens for the possibility of a successful analysis at the neonatal stage in the mouse. It also provides a link between a large body of studies in the adult cat and the modern approaches investigating electrophysiology and anatomy of neurons expressing selective transcriptional factors identified and manipulated by genetic tools. However, this may raise the question on whether the interneurons and their connections investigated in the present study are fully developed at that early postnatal stage. Therefore it is important to point out that Razgado and Quevedo (2004) has demonstrated that low threshold ankle extensor afferents indeed cause resetting to extension in the neonatal rat preparation. In order to resolve the questions we have addressed in this study it seems that the future requires extensive collaborations between classical researchers using “adult animals in vivo preparations” and the new generation of neuroscientists working with transgenic animals, and using genetic tools. That is the type of collaboration that was certainly fostered by Tom Jessell and his colleagues.

AUTHOR CONTRIBUTIONS

Hultborn, H., Quevedo, J.N. and Stecina, K., contributed to the conception and design of the study. Material preparation, data collection, and analysis were performed by **Domínguez-Rodríguez L. E., Stecina K., García-Ramírez D. L., Mena-Avila E., Milla-Cruz J. J., Martínez-Silva L., Zhang M., Hultborn H. and Quevedo J. N.** The first draft of the manuscript was written by **Domínguez-Rodríguez L.E.** and all authors commented on the previous versions of the manuscript. All authors read and approved the final manuscript.

DECLARATION OF INTEREST

None

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LEGENDS

Figure 1. Two layers model of locomotor CPG based on Graham Brown's "Half-Center" hypothesis, and the extensor gr I excitatory pathway to extensors during locomotion. A. The schematic outlines the half-center model of the CPG receiving projections from ipsilateral (*i*FRAs) and contralateral (*co*FRAs) flexor reflex afferents (Brown TG, 1911, 1914; Jankowska et al., 1967 a, b). It also illustrates the two layers architecture of locomotor CPG with the rhythmic generator (RG) located in the ventral horn of the spinal cord and the pattern formation layer (PF) sending projections onto motoneurons. First-order interneurons are located in the intermediate nucleus. **B.** During fictive locomotion (evoked either from MLR, or by L-DOPA) the stimulation of extensor gr I afferents facilitates strong excitation (pathways: *b* and *c*) which overcomes the non-reciprocal inhibition seen at rest (pathway: *a*) (Gossard et al., 1994; McCrea et al., 1995). It has been

hypothesized that the polysynaptic pathway (*c*) is related to the resetting of the locomotor rhythm with second and last-order interneurons (2, 3 in *c*) belonging to the extensor half-center, and then they should receive excitation from *coFRA* (Conway et al., 1987; Gossard et al., 1994) and descending excitation from MLR. Finally, in A it is shown how extensor gr I afferents have access to the extensor half-center of the CPG through this polysynaptic pathway (*c* in B).

Figure 2. Polysynaptic gr I evoked EPSPs in an extensor motoneuron during L-DOPA-

induced locomotion. A. Intracellular recording (IC) from a GS motoneuron (with QX-314 in the microelectrode to block action potentials). Second trace, IC recording of locomotor drive potentials (LDPs). Upper traces, vertically displayed IC recordings to visualize the EPSPs evoked by the stimulation of 5 pulses to *iPl* nerve at 2 xT (given at 2 Hz). EPSPs during the flexor phase are displayed in red and EPSPs during the extensor phase in blue. Bottom traces, rectified and integrated ENG recording from the extensor *iSmAB* (blue) and the flexor *iTA* (red) nerves. **B.** Phase-dependent modulation of polysynaptic EPSPs. **Bi.** From the top to the bottom: 1) Averages of extensor gr I evoked EPSPs during flexion (red) and during extension (blue). They were superimposed with the corresponding Em both during flexion (flex) and extension (ext); 2) the difference (subtraction) between the response during the flexor and extensor phase; 3) the cord dorsum potential (CDP) recording of the afferent volley evoked by 1 pulse to *iPl* at 2 xT. The central latency is measured between the beginning of the afferent volley (arrow) and the beginning of EPSPs (dotted line). The latency of 3.1 ms suggests 3 interneurons interposed between extensor gr I afferents and the recorded motoneuron. **Bii.** Same format as in Bi, but in this case EPSPs were evoked by a train of 5 pulses to *iPl* at 2 xT. The bottom trace shows the extracellular recording (EC) after leaving the GS motoneuron. Note the higher amplitude of gr I EPSPs evoked during flexion than during extension. **C.** Stimulation of extensor gr I afferents promotes the onset of the extensor phase. DC recordings from a GS MN under control condition without stimulation of extensor gr I afferents (upper trace,) and during stimulation (lower trace). The stimulation of *iPl* (5 pulses, 2 xT) shortens transition time (p value of t-test = 0.0004). This suggests that stimulation of *iPl* promoted the transition to initiate the extensor phase, as shown by the examples in A by faster transitions indicated by black arrows, as compared to the step marked by a star where stimulus train was given before spontaneous phase transition (see text). When stimulus trains were given at

2 Hz the transition time was significantly shorter than during the control period with no train stimulation to *iPl*, as observed in the right histogram.

Figure 3. Expression of disynaptic EPSPs evoked by stimulation of extensor gr I afferents during L-DOPA-induced locomotion. Intracellular recording (IC) from a GS motoneuron with QX-314 in the microelectrode to block action potentials. **A.** Upper traces, EPSPs evoked by single stimuli to the *iGS* nerve (2 xT). Second trace, recording of the LDPs for over 40 s (three steps). The labeled EPSPs correspond to the upper-most part individual EPSPs. Stars mark EPSPs evoked during flexor (red) and triangles during extensor (blue) phase of locomotion. Lower traces, locomotor episode represented by rectified and integrated ENG of ipsilateral *iSmAB* and *iTA*. **B.** Average of monosynaptic EPSPs occurred during flexion with a latency of 0.6 ms. The central latency is estimated from the beginning of the afferent volley (arrow) to the beginning of EPSPs (dotted line). **C.** Same format as in B, but during extension. Note that after the monosynaptic peak there is an emergence of EPSPs with a disynaptic latency (1.4 ms). This disynaptic pathway expressed during extension was previously described by McCrea et al. (1995) and Angel et al. (1996) during MLR-locomotion.

Figure 4. First-order interneuron with monosynaptic connection from extensor gr I afferents rhythmically activated during extension. **A.** Upper trace, extracellular recording from an interneuron localized at 2.5 mm deep from surface, activated by stimulation of the FDHL nerve (2 xT, 4 pulses) with a latency of 0.9 ms, suggesting that it receives monosynaptic connections from *iFDHL* gr I afferents. Lower trace, CDP recordings of the stimulated nerve. Here, and in the following figures (5-7), the central latency is measured between the beginning of the afferent volley (arrow) and the onset of the extracellular spikes (dotted line). **B.** From top to bottom, rhythmic firing of the same interneuron during L-DOPA-induced locomotion. Rectified and integrated ENG of *iGS* and *iFDHL* extensor (blue) and *iTA* flexor (red) nerves. **C.** The sketch illustrates the polysynaptic pathway from extensor gr I afferents during locomotion. The filled blue circle represents the first- order interneuron, indicated by the microelectrode. **D.** From top to bottom, extracellular recording of the same interneuron activated with a long-latency by the

stimulation of “flexor reflex afferents” in the *coSur* (50 xT, 300Hz), raw ENG recordings from ipsilateral extensor *iGS* (blue), and the ipsilateral flexor *iTA* (red) nerves.

Figure 5. Interneuron with disynaptic activation from extensor gr I afferents and rhythmically active during extension. **A.** Upper trace, extracellular response of an interneuron to stimulation of extensor gr I afferents from *iGS* nerve (2 xT, 3 pulses), located 2.3 mm deep from surface. The latency of 1.3 ms from incoming volley (CDP; lower trace) suggests a disynaptic connection from the gr I *iGS* afferents. **B.** From top to bottom, extracellular recording from the same interneuron firing during the extensor phase of L-DOPA-induced locomotion, rectified and integrated ENGs of *iGS* and *iFDHL* extensor (blue) and *iTA* flexor (red) nerves. **C.** The sketch illustrates the polysynaptic pathway from extensor gr I afferents expressed during locomotion. The filled blue circle represents the second-order interneuron recorded. **D.** From top to bottom, extracellular recording of the long-latency activation of this interneuron by the stimulation of *coSur* (50 xT, 300Hz), raw ENG recordings from *iFDHL* (blue) and *iTA* (red) nerves. **E.** From top to bottom, extracellular activity of this interneuron follows the resetting to extension in response of a long stimulus train to the *iPl* nerve (2 xT, 300 Hz) during flexion, and rectified and integrated ENG recordings from *iGS* and *iFDHL* extensor (blue), and *iTA* flexor (red) nerves. Bottom trace show the period of stimulation (stimulus tags).

Figure 6. Last-order interneuron receiving di- and trisynaptic excitation from extensor gr I afferents. **A.** Extracellular response evoked by stimulation of gr I fibers from *iPl* at 2 xT (upper trace, Ai) and from *iSmAB* at 5 xT (which would include some gr II afferents as well) (upper trace, Aii) of an interneuron localized in the medial part of the ventral horn, at a depth of 4.2 mm from the surface. In Ai stimulation of the *Pl* nerve evoked spikes with latencies that vary between 1.3-2.8 ms, as measured from the onset of the incoming volley. This suggests both disynaptic and trisynaptic excitation from gr I extensor afferents. In Aii the shortest latency was 1.2 ms suggesting disynaptic projections. Lower traces in Ai and Aii show the CDP recordings. **B.** From top to bottom, extracellular recording of the rhythmic activity of this interneuron during the extensor phase of L-DOPA-induced locomotion, and rectified ENGs of *iSmAB* and *iPl* extensor (blue) and *iTA* flexor (red) nerves. The gray bars show that the interneuron fires during the brief and robust

extensor activity. **C.** Interneuronal discharge evoked by micro-stimulation (15 μ A) of the extensor motor nucleus. The measured latency from the onset of the stimulus artefact to the onset of the spike was 0.5 ms indicating antidromic activation from the motor nucleus. **D.** Spike-triggered averaging (STA) analysis shows a response in *iSmAB* ENG associated with the spike. Measured latency of 1.5 ms suggests a monosynaptic projection from this interneuron to *SmAB* motoneurons. **E.** The sketch illustrates the polysynaptic pathway from extensor gr I afferents during locomotion. As it is indicated by the microelectrode, the filled blue circle represents this interneuron receiving a disynaptic and trisynaptic projections from extensor gr I afferents, and making monosynaptic contact with extensor motoneurons.

Figure 7. Last-order interneuron antidromically activated from the GS motor nucleus with polysynaptic activation from extensor gr I afferents. **Ai and Aii.** Upper traces, extracellular discharge of an interneuron evoked by the stimulation of extensor gr I afferents from *iGS* and *iSmAB* (2 xT, 3 pulses) with latencies around 5- 6 ms (vertical dotted lines), suggesting a polysynaptic connection from extensor gr I afferents. Lower traces, CDP recordings of the stimulated nerves. **B.** Upper trace, extracellular recording from the same interneuron. Lower traces, rectified and integrated ENG of *iFDHL* (second trace, blue) and *iTA* (third trace, red) nerves. The interneuron was rhythmically firing during the extensor phase (blue) of L-DOPA-induced locomotion. Note that the activity in the FDHL nerve exhibits two peaks, likely corresponding to flexor hallucis longus and digitorum longus, respectively. **C.** Extracellular spikes evoked by micro-stimulation (20 μ A) in the GS motor nucleus. The measured latency from the end of the stimulus artefact to the onset of the spike was 0.43 ms indicating antidromic activation from the GS motor pool. **D.** This interneuron represents the last-order interneuron receiving polysynaptic excitation from extensor gr I afferents, as indicated by the microelectrode outside the filled blue circle. The interneuron likely makes monosynaptic contact with GS motoneurons, since it was antidromically activated from the GS motor pool.

Figure 8. Location of interneurons fulfilling the criteria to belong to the extensor gr I polysynaptic excitatory pathway. Most of the interneurons were recorded in L6 and rostral L7 segments (see further in the text), and their location was therefore fitted to the outline of a

transversal section from the L6 segment, as illustrated by Rexed (1954). **A.** The schematic illustrates the position of all 62 interneurons recorded during fictive locomotion (and/or scratch). All of them exhibit rhythmic spontaneous firing during the extensor phase (as part of the inclusion criteria). Blue circles represent interneurons receiving monosynaptic projections from extensor gr I afferents (n=14). Green circles are interneurons activated mono- and disynaptically (n=4). Yellow circles are interneurons with disynaptic activation (n=8). Orange circles indicate interneurons with a mixture of responses: mono-, di- and polysynaptic (n=23). Red circles are interneurons which receive only polysynaptic excitation from extensor gr I afferents (n=13). **B.** Here is shown all the interneurons with a monosynaptic extensor gr I component (even if they also received additional di- or polysynaptic excitation); see further in text. **C.** Location of the four interneurons illustrated in Figs. 4-7.

Figure 9. Two possible hypotheses to explain how the excitatory polysynaptic pathway from extensor gr I afferents to extensor motoneurons relates to the locomotor extensor half-center and CPG. **A.** Stimulation of extensor gr I afferents at rest produces non-reciprocal inhibition on extensor motoneurons mediated by pathway *a*. During locomotion, there is a facilitation of the interneurons of two excitatory pathways: one that mediates disynaptic EPSPs on extensor motoneurons (*b*), and the other mediating the polysynaptic EPSPs (*c*) related to the resetting of the locomotor rhythm, which could imply that this path is actually part of the extensor half-center of the CPG (light blue). This polysynaptic path is composed by three population of interneurons between extensor gr I afferents and extensor motoneurons. **B.** Another possible explanation could be that this polysynaptic gr I excitatory pathway is not part of the CPG, but facilitated both by the extensor half-center of the CPG; (light blue) *and* the flexor counterpart (light pink).

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FIGURES

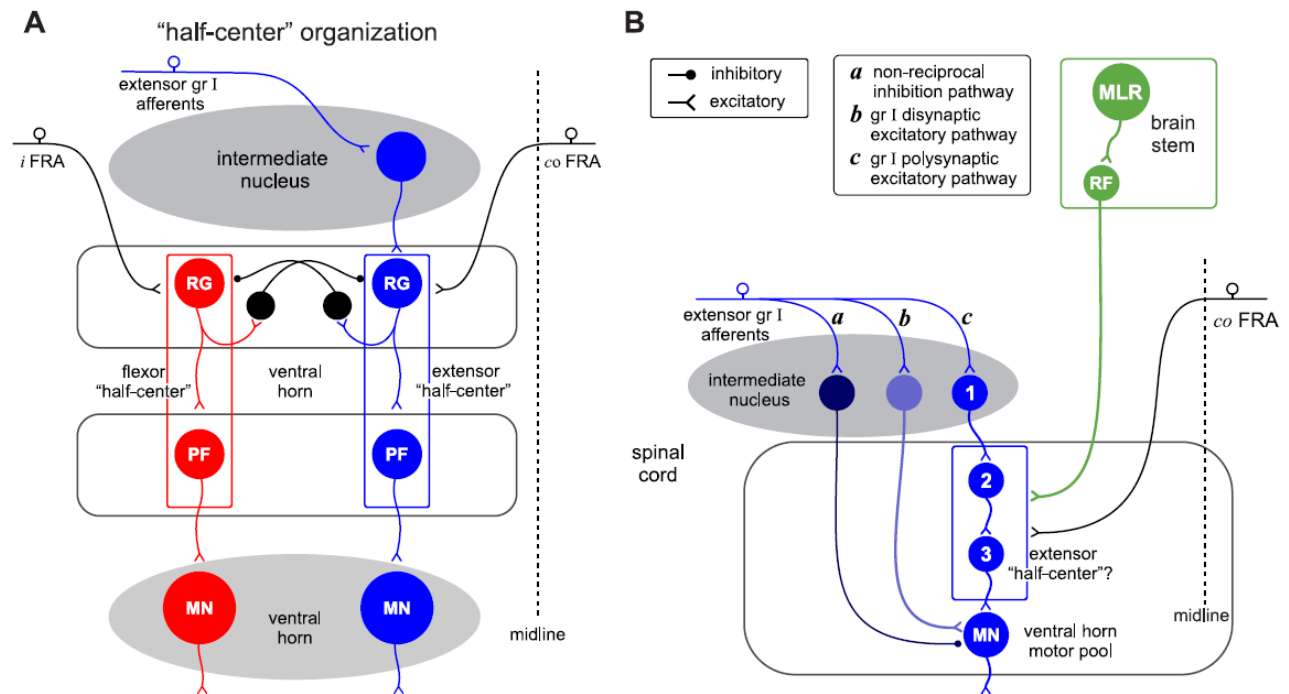


Figure 1. Two layers model of locomotor CPG based on Graham Brown’s “Half-Center” hypothesis, and the extensor gr I excitatory pathway to extensors during locomotion

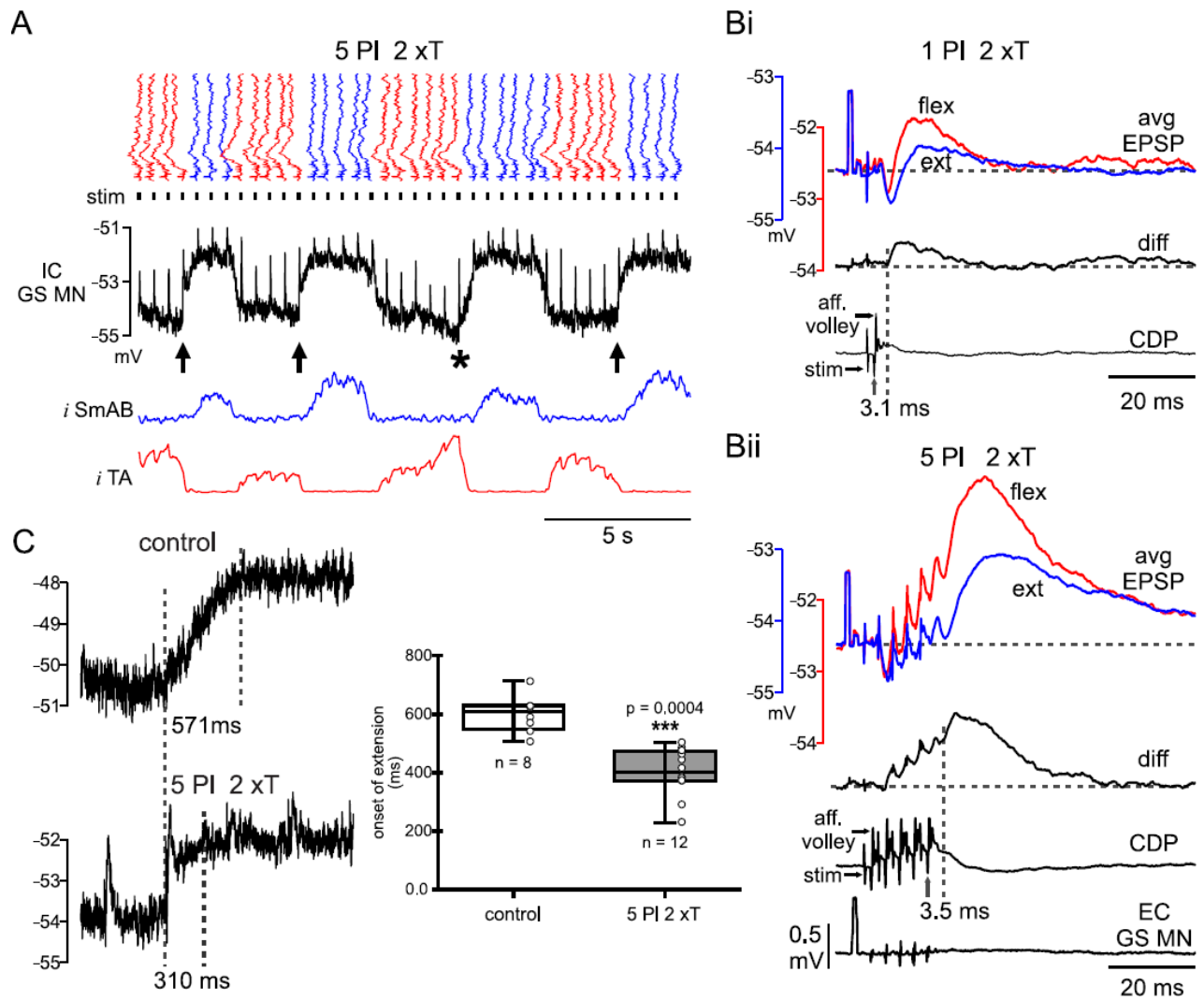


Figure 2. Polysynaptic gr I evoked EPSPs in an extensor motoneuron during L-DOPA-induced locomotion.

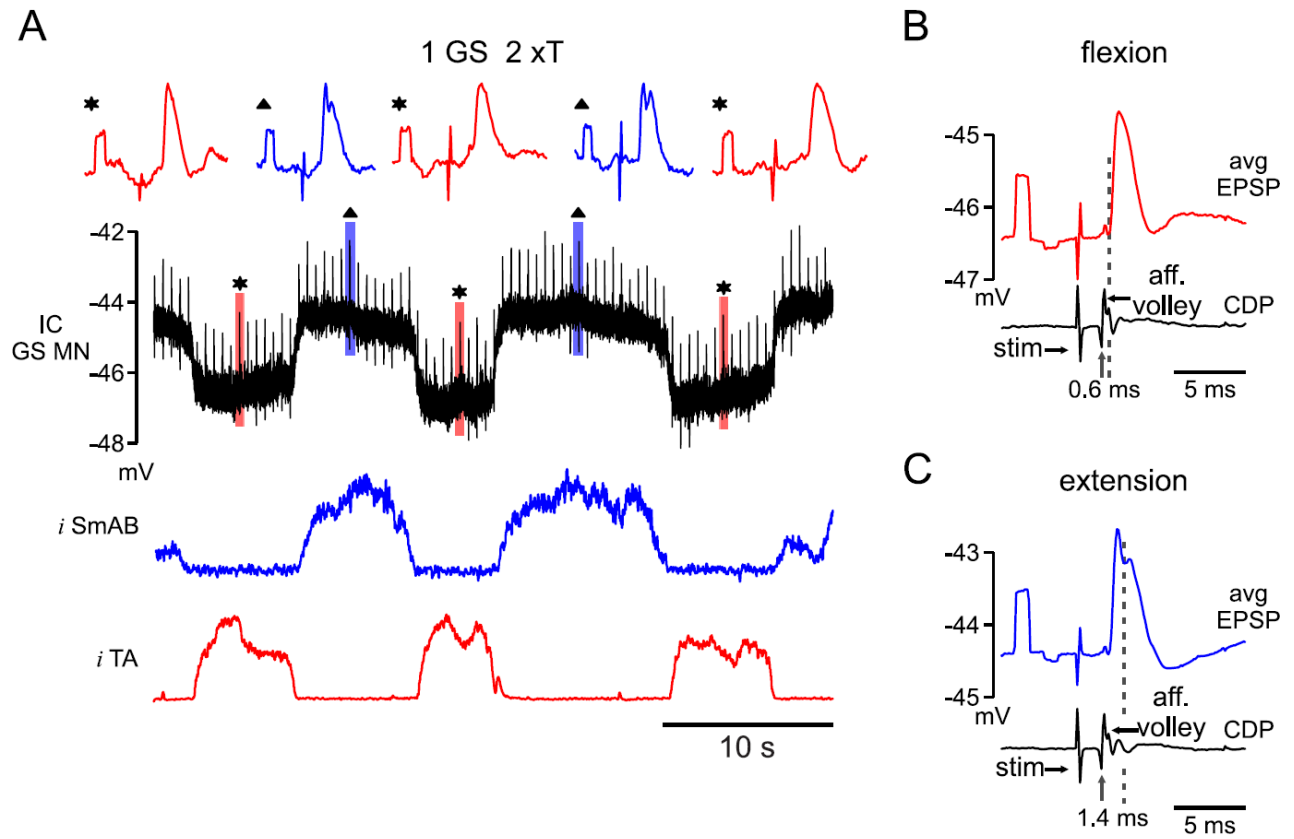


Figure 3. Expression of disynaptic EPSPs evoked by stimulation of extensor gr I afferents during L-DOPA-induced locomotion.

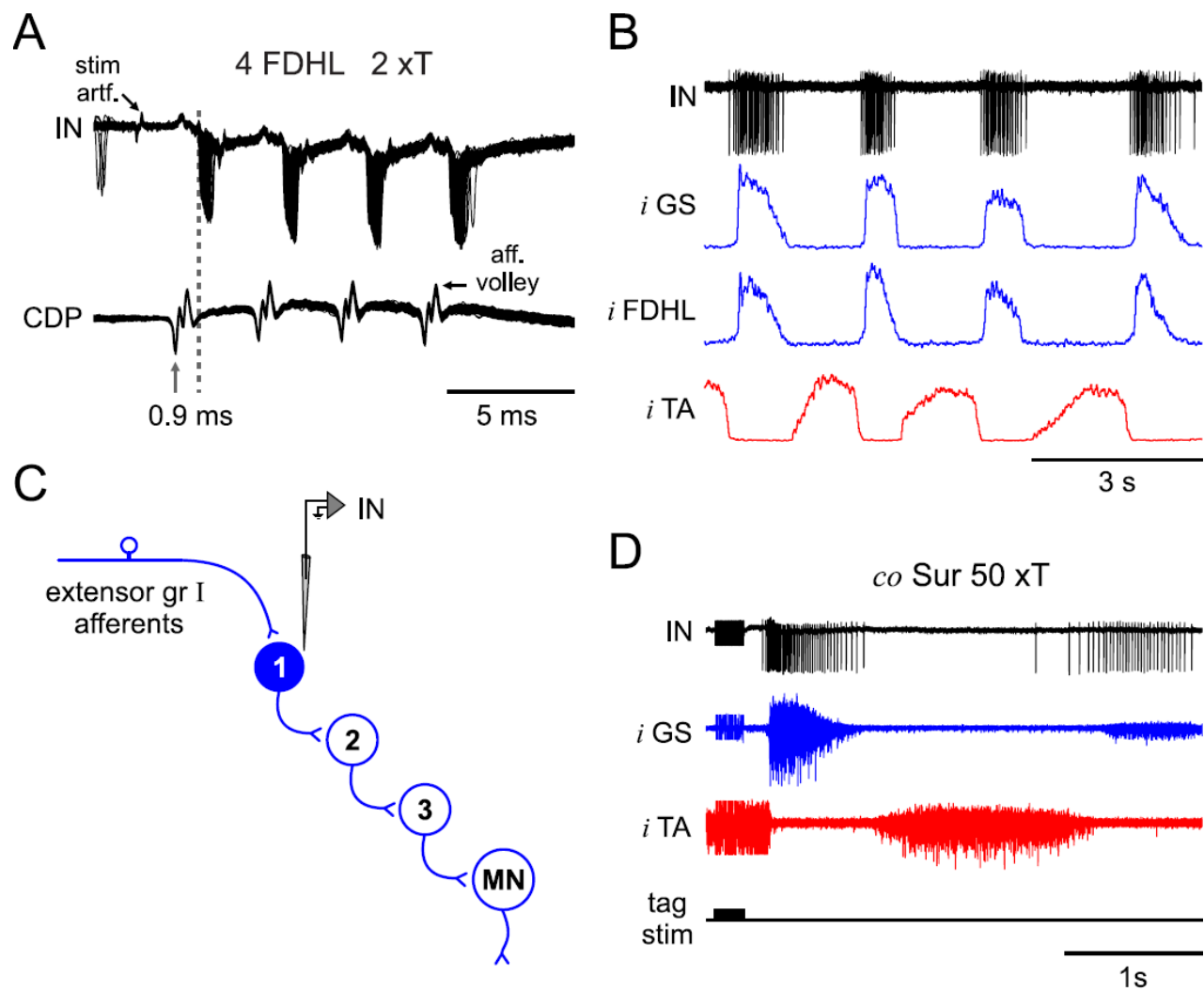


Fig. 4. First-order interneuron with monosynaptic connection from extensor gr I afferents rhythmically activated during extension

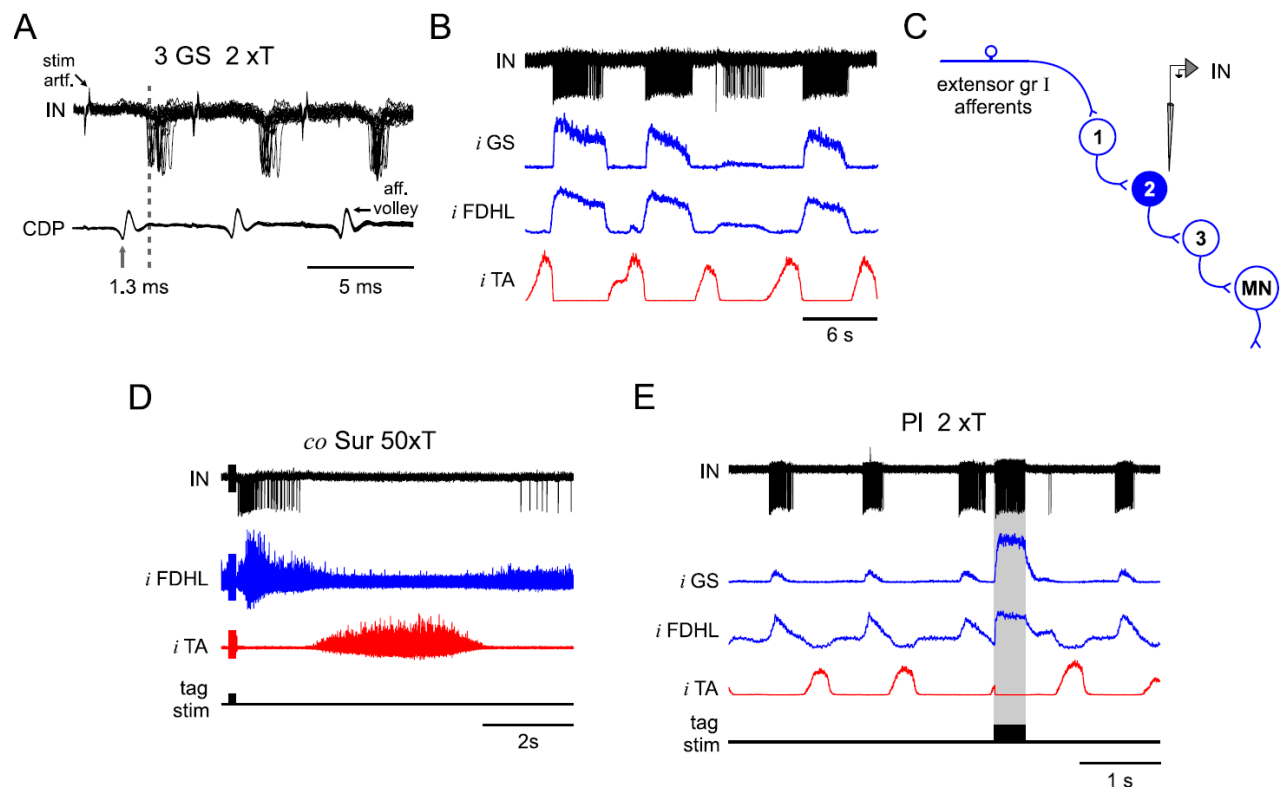


Figure 5. Interneuron with disynaptic activation from extensor gr I afferents and rhythmically active during extension

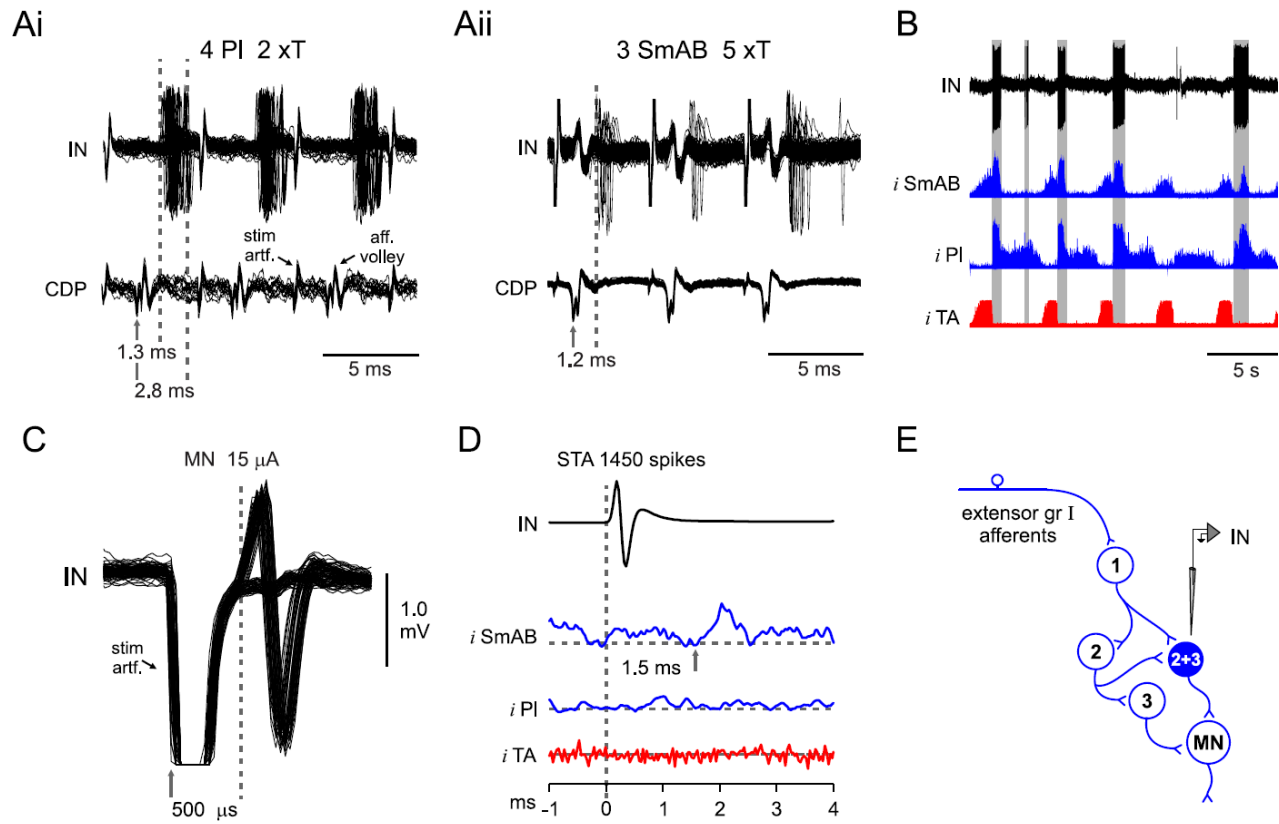


Figure 6. Last-order interneuron receiving di- and trisynaptic excitation from extensor gr I afferents.

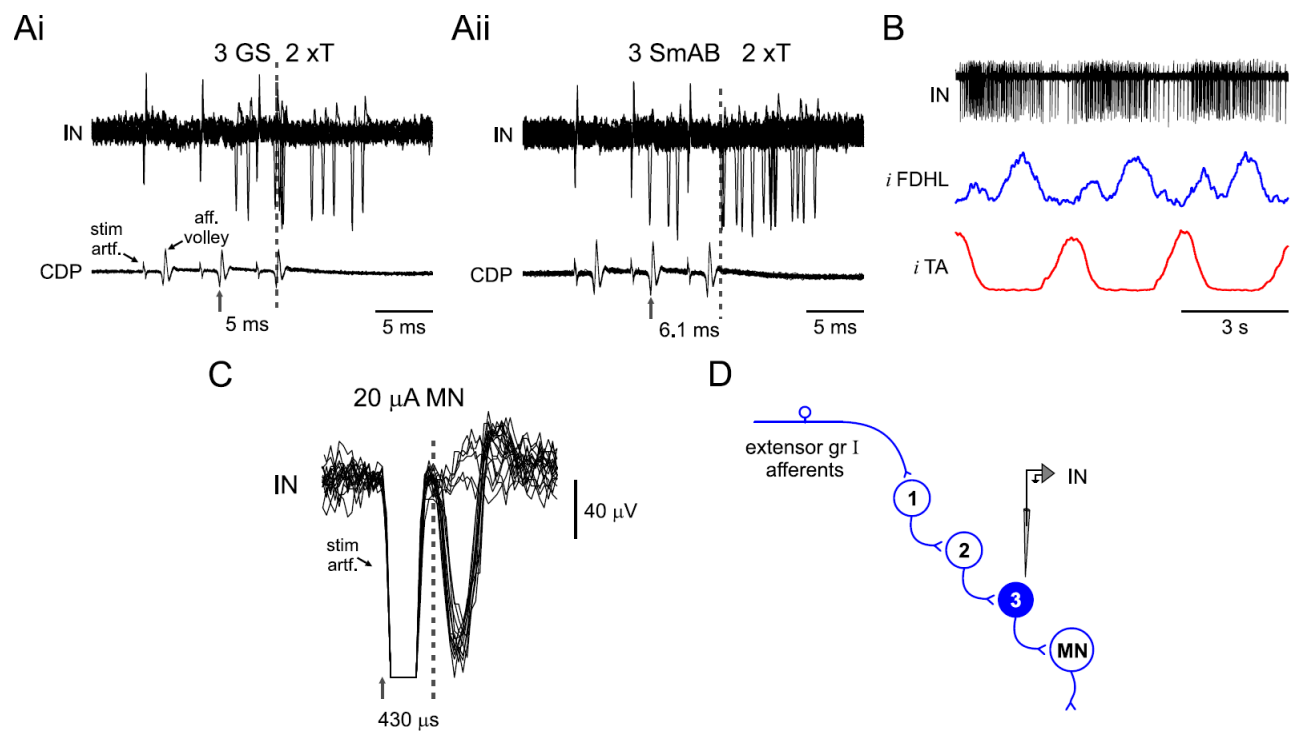


Figure 7. Last-order interneuron antidromically activated from the GS motor nucleus with polysynaptic activation from extensor gr I afferents

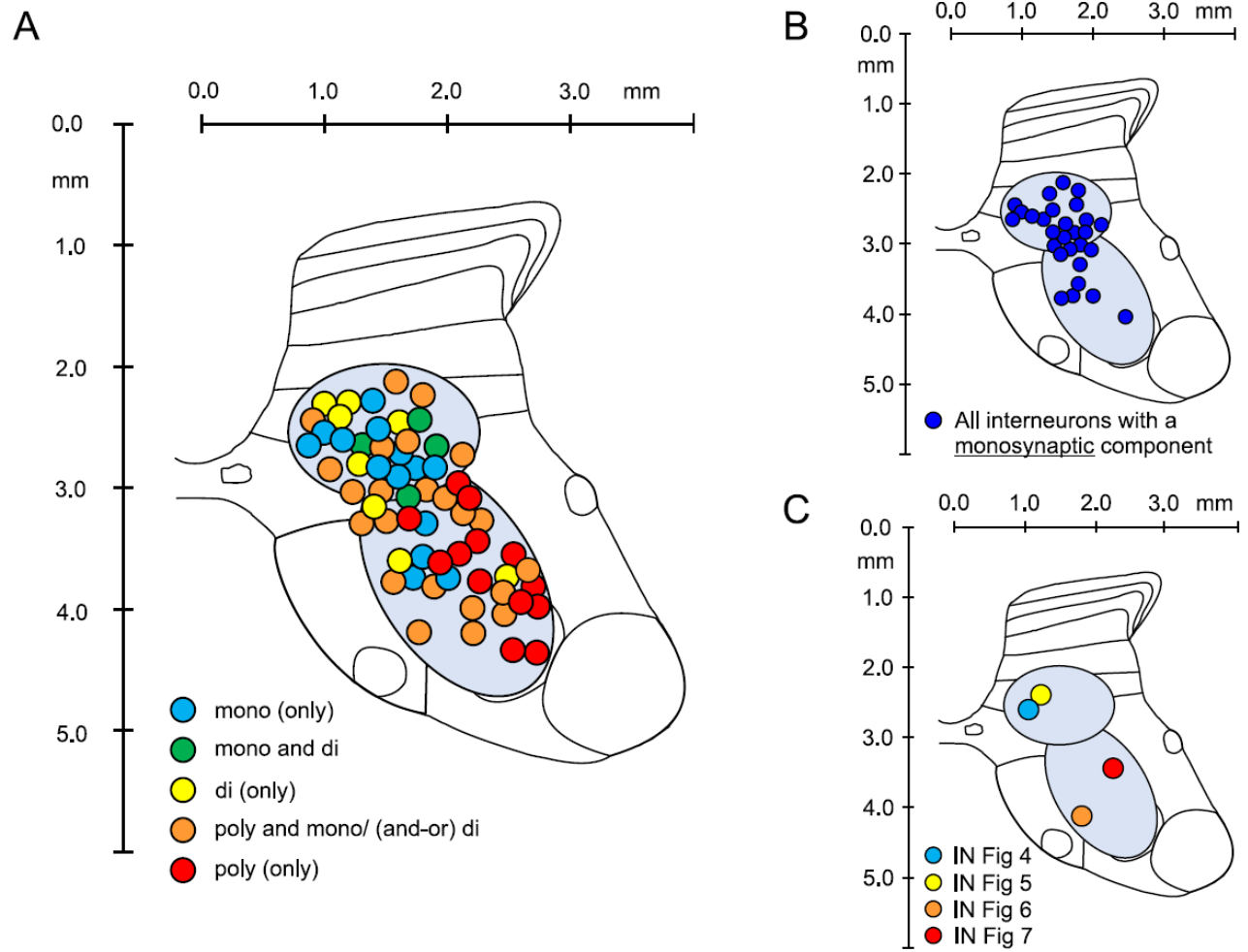


Figure 8. Location of interneurons fulfilling the criteria to belong to the extensor gr I polysynaptic excitatory pathway.

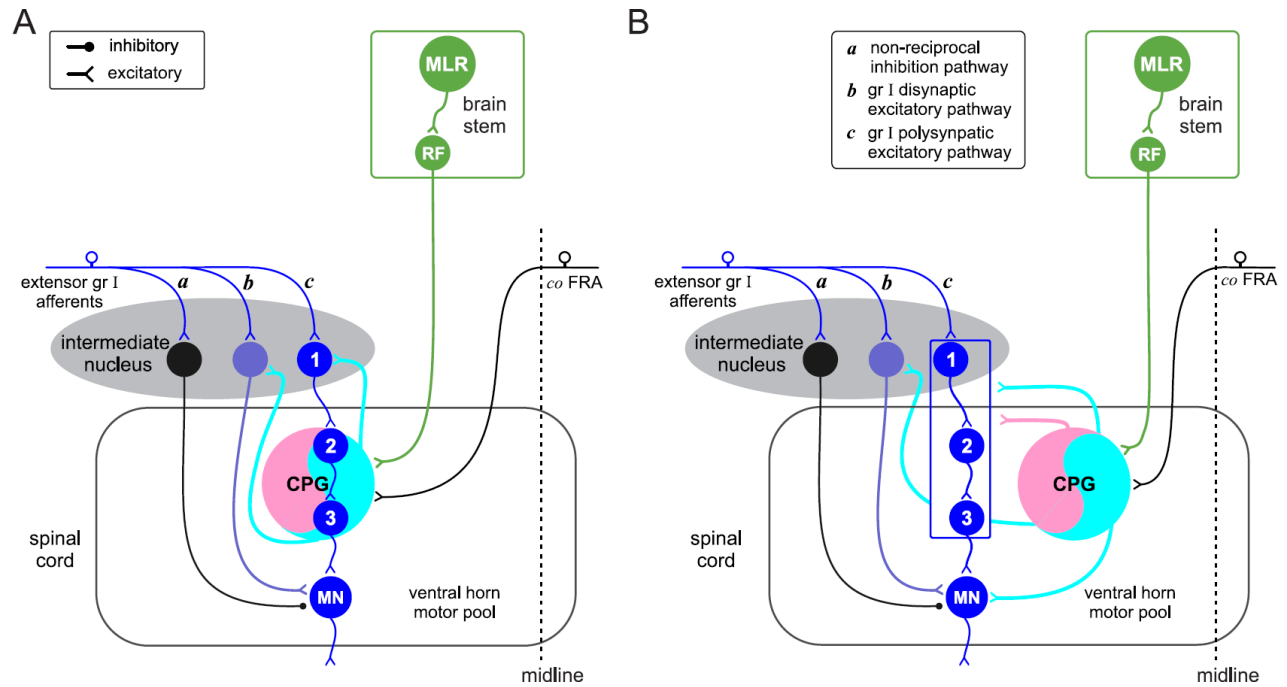


Figure 9. Two possible hypotheses to explain how the excitatory polysynaptic pathway from extensor gr I afferents to extensor motoneurons relates to the locomotor extensor half-center and CPG.